AP PRECALCULUS

Unit 1 • Lesson 1.13 practice • Answer key

Lesson 1.13 • Function Model Selection and Assumption Articulation

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

Data with equally spaced inputs have first differences $4.9,\ 5.1,\ 5.0,\ 4.9$. Which model is most appropriate?

(A) Linear

(B) Quadratic

(C) Cubic

(D) Rational

Solution

Answer: (A). The rate of change is roughly constant.

Question 2

Data with equally spaced inputs have second differences $-6.1,\ -5.9,\ -6.0$. Which model is most appropriate?

(A) Linear

(B) Quadratic

(C) Cubic

(D) Constant

Solution

Answer: (B). Roughly constant second differences indicate a quadratic model.

Question 3

A company's average cost per item is $A(n) = \dfrac{500 + 4n}{n}$ dollars, for n items. What does $\lim_{n \to \infty} A(n)$ tell you?

(A) The fixed cost is \$500.

(B) The average cost becomes 0.

(C) The average cost per item approaches \$4.

(D) The cost per item increases without bound.

Solution

Answer: (C). $A(n) = \frac{500}{n} + 4 \to 4$: the fixed cost is spread over more and more items.

Question 4

A quadratic model $P(t)$ gives the number of fish in a pond after t months; it has a maximum, then decreases. Which is the most reasonable restriction on the domain?

(A) All real numbers t

(B) $t < 0$

(C) Only integer values of P

(D) $t \geq 0$ and only while $P(t) \geq 0$

Solution

Answer: (D). Time starts at 0 and a population cannot be negative.

Question 5 (Free response)

A phone plan costs \$20 per month, which includes 100 texts. Each additional text costs \$0.10.

(a) Write a piecewise function $C(n)$ for the monthly cost of sending n texts.
(b) Find $C(250)$ and interpret it.
(c) For $n > 100$, the average cost per text is $A(n) = \dfrac{C(n)}{n}$. Find $\lim\limits_{n\to\infty} A(n)$ and interpret it.

Solution

(a) $C(n) = \begin{cases} 20, & 0 \le n \le 100 \\ 20 + 0.10(n - 100), & n > 100 \end{cases}$

(b) $C(250) = 20 + 0.10(150) = 35$: sending 250 texts in a month costs \$35.

(c) For $n > 100$, $A(n) = \dfrac{10 + 0.10n}{n} = \dfrac{10}{n} + 0.10 \to 0.10$. With many texts, the average cost per text approaches \$0.10, the price of an extra text.